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Spectral Line Shapes

Gaussian and Lorentzian functions in magnetic resonance

Spectroscopic experiments may vary considerably in their details yet they all involve the same fundamental phenomenon: transitions are induced between discrete energy levels in which the system under consideration exists, by irradiating the system with electromagnetic radiation of the appropriate frequency so as to satisfy the Bohr frequency condition, $\Delta E = \hbar\omega$ (ΔE is the energy difference between any two energy levels, $\hbar$ is Planck's contant, and ω is the frequency of the transition-inducing electromagnetic field).

Selection rules determine which of the many possible transitions are likely to be observed, and transition probabilities determine the relative intensities of the transitions. Analysis of the resulting spectrum then, i.e., extraction of the appropriate spectroscopic parameters such as transition frequencies, intensities, line shapes, splitting of bands, etc., allows the spectroscopist or chemist to deduce information about the system that is giving rise to the particular spectrum studied. The analytical spectroscopists and chemists have hitherto paid attention mostly to the band frequencies. The widespread use of ir group frequencies and nmr chemical shifts in structure determination problems attest to this fact.

Yet the band shape itself and the various statistical parameters that can be derived from the band shape (e.g., areas, moments, half-widths, etc.) may contain much significant information. For example, vapor-phase measurements of ir band intensities make possible in principle the determination of the dipole moment change associated with a given normal vibration as well as the effect of vibrational motion on the polarization of molecules. Condensed-phase measurements of ir band intensities may provide information on the nature of intermolecular forces (*1*). Analysis of line shapes in magnetic resonance of solids can yield information on the distribution of the species undergoing resonance (random vs. ordered distribution) and on the strength of their interaction with neighboring species (*2–5*). It is natural then that spectral line shapes would receive increasing attention from the practicing spectroscopists. Of course, often band overlap complicates the problem and makes extremely difficult the analysis of spectral line shapes. But for nonoverlapping or resolvable bands the first step is the determination of the line shape function itself. This not only may be significant information in itself, but it also may allow the calculation of the various statistical parameters used to characterize the band itself from the simple measurements of line width and amplitude.

Two very commonly encountered line shapes in various branches of molecular spectroscopy are the Gaussian and the Lorentzian. Where collision broadening is the main factor that determines the band half-width (as for example in the vapor-phase ir spectra and high resolution nmr) the resulting bands are Lorentzian (*1*, *6*). On the other hand, rigid systems in which the relative orientations and positions of randomly distributed and interacting species do not change with time give rise to Gaussian lines (*2*, *6*). It is not our intention to examine here the factors that determine the line shapes for the various spectroscopic techniques. Rather, we intend to examine the properties of these two very commonly encountered line shapes, the Gaussian and the Lorentzian. We will specialize our discussion somewhat in terms of nmr and esr spectra, yet what is written here is generally applicable to all spectroscopy where these two band shapes are encountered.

The Resonance Phenomenon

In magnetic resonance experiments one observes transitions between the Zeeman levels of electrons (esr) or nuclear Zeeman levels of nuclei (nmr). A stationary magnetic field H removes the degeneracy associated with electron or nuclear spins by causing them to assume definite orientations with respect to the field direction. Since electrons and nuclei can be viewed as microscopic magnetic dipoles, they interact with the applied external field, the energy of interaction E being,

$$E = -\boldsymbol{\mu}\cdot\mathbf{H} = -\mu H \cos\theta \tag{1}$$

where μ is the magnetic dipole moment, H is the externally applied field, and θ is the angle between μ and H. For situations where the spin is $^1/_2\,\hbar$ (electrons, protons, ^{19}F nuclei, ^{31}P nuclei, etc.) only two angles are admissible according to the laws of quantum mechanics, 0° and 180°. Therefore, the energy difference between the two levels is

$$\Delta E = 2\mu H \tag{2}$$

and as a result transitions between these two levels will occur when the system is irradiated with energy of frequency

$$\omega = 2\mu H/\hbar = \gamma H \tag{3}$$

This is the famous Larmor equation. It states that the frequency at which resonance will occur is proportional to the applied magnetic field responsible for the removal of the spin degeneracy, the proportionality constant being the magnetogyric ratio of the species undergoing resonance. Normally, in magnetic resonance experiments the frequency is kept constant and the field is swept. The resulting spectra then are traces of the absorption versus the field H, or by using the appropriate experimental setup the first or even the second derivative of the absorption may be recorded. In fact, this is standard practice in esr and wide-line nmr.

The Gaussian and Lorentzian Functions

Absorption occurs not at one single frequency, but rather over a range of frequencies resulting in broadened lines or bands. This, of course, is characteristic of all spectroscopic techniques although the specific factors that cause the broadening may be different.

The most commonly encountered line shapes are the Gaussian and the Lorentzian, eqns. (4) and (5), respectively.

$$F(H - H_0) = \frac{\sqrt{\ln 2}}{\sqrt{\pi}\delta} \exp\left[-(H - H_0)^2 \ln 2/\delta^2\right] \quad (4)$$

$$F(H - H_0) = \frac{\delta}{\pi} \cdot \frac{1}{\delta^2 + (H - H_0)^2} \quad (5)$$

where $F(H - H_0)$ is the amplitude of the function, δ is one-half of the width of the function at half-maximum intensity, and $(H - H_0)$ is the value of the field measured from the center of the resonance (Table 1). A plot of these two functions is shown in Figure 1a. It is assumed that both functions have the same half-line width. Two significant differences between these two line shapes are evident: the Lorentzian is a sharper, narrower function toward the center; also, it falls off in the wings not as rapidly as the Gaussian. At a distance from the center equal to three half-line widths, the amplitude of the Gaussian is essentially zero, while at the same point the amplitude of the Lorentzian is approximately 10% of its maximum amplitude (Table 2).

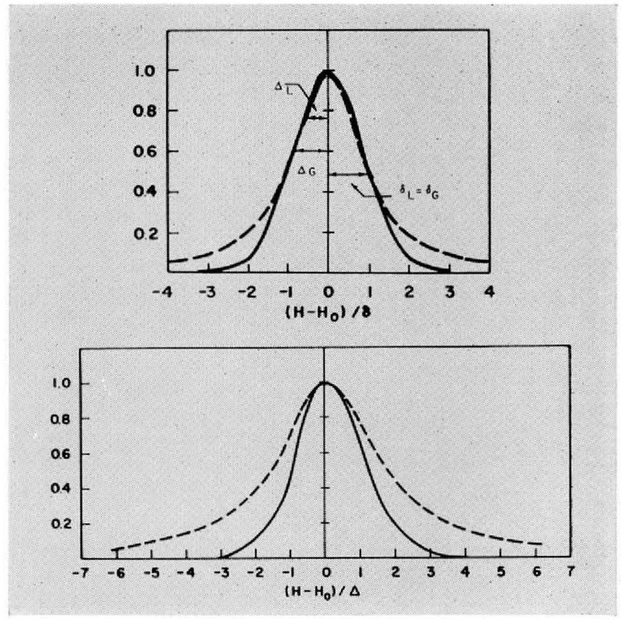

Figure 1. Zero derivative curves for Gaussian (solid line) and Lorentzian (broken line) shape function; (a) curves have the same half-line width at half-maximum intensity; (b) curves have the same peak-to-peak line width.

It is also clear from Figure 1a that although the half-line widths at half-maximum intensity are the same for the two functions, the point at which the slope is maximum for each curve is very different. Close examination of the curves indicates that for the Gaussian the maximum slope of the absorption is very close to the half-intensity mark, while for a Lorentzian the maximum slope is at about the three-quarter mark of the maximum intensity.

If each of expressions (4) and (5) is differentiated twice, set equal to zero and solved for $(H - H_0)$, the values of the field $(H - H_0)$ are obtained at which the slope is maximum. These values are, respectively, for the Gaussian and the Lorentzian, $\delta/\sqrt{2 \ln 2}$ and $\delta\sqrt{3}$. These are the so-called peak-to-peak line widths

Table 1. Analytical Expressions for Gaussian and Lorentzian Functions

	Gaussian	Lorentzian
$F(H - H_0)$	$\frac{1}{\sqrt{2\pi}\Delta} \exp[-(H - H_0)^2/2\Delta^2]$	$\frac{\sqrt{3}\Delta}{\pi} \cdot \frac{1}{3\Delta^2 + (H - H_0)^2}$
$F(H - H_0)$	$\frac{\sqrt{\ln 2}}{\sqrt{\pi}\delta} \exp[-(H - H_0)^2 \ln 2/\delta^2]$	$\frac{\delta}{\pi} \cdot \frac{1}{\delta^2 + (H - H_0)^2}$
δ	$\sqrt{2 \ln 2}\Delta = 1.77\ \Delta$	$\sqrt{3}\Delta = 1.732\ \Delta$
$F(H - H_0)_{max}$	$1/\sqrt{2\pi}\Delta$	$1/\sqrt{3}\pi\Delta$
$F'(H - H_0)$	$-\frac{(H - H_0)}{\sqrt{2\pi}\Delta^3} \exp[-(H - H_0)^2/2\Delta^2]$	$-\frac{2\sqrt{3}\Delta(H - H_0)}{\pi} \cdot \left[\frac{1}{3\Delta^2 + (H - H_0)^2}\right]^2$
$F'(H - H_0)_{max}$	$1/\sqrt{2\pi e}\Delta^2$	$\sqrt{3}/8\pi\Delta^2$
$F'(H - H_0)_\delta$	$\ln 2/\sqrt{4\pi}\Delta^2$	$1/6\pi\Delta^2$
$F''(H - H_0)$	$\frac{1}{\sqrt{2\pi}\Delta^3}\left[\frac{(H - H_0)^2}{\Delta^2} - 1\right] \exp[-(H - H_0)^2/2\Delta^2]$	$\frac{6\sqrt{3}\Delta}{\pi} \cdot \frac{(H - H_0)^2 - \Delta^2}{[3\Delta^2 + (H - H_0)^2]^3}$
$F''(H - H_0)_{max}$	$-1/\sqrt{2\pi}\Delta^3$	$-2/\sqrt{3}/9\pi\Delta^3$
$F''(H - H_0)_{\sqrt{3}\Delta}$	$2/\sqrt{2\pi e^3}\Delta^3$	$\sqrt{3}/18\pi\Delta^3$
K[a]	1.062	1.572
K′[b]	0.517	1.812
Second moment	Δ^2	∞
Fourth moment	$3\Delta^4$	∞

[a] See eqn. (5).
[b] See eqn. (6).

Table 2. Numerical Values for Absorption and Derivatives of Gaussian and Lorentzian Functions

		Gaussian			Lorentzian			
Δ	δ	$F(H - H_0)$	$F'(H - H_0)$	$F''(H - H_0)$	δ	$P(H - H_0)$	$F'(H - H_0)$	$F''(H - H)$
0	0	1.0	0	−1.0	0	1.0	0.0	−1.0
0.578	0.491	0.846	0.60	−0.562	0.334	0.895	0.825	−0.478
0.848	0.719	0.698	0.855	−0.195	0.490	0.806	0.983	−0.147
1.00	0.848	0.606	1.00	0.0	0.578	0.750	1.00	0.00
1.155	0.981	0.514	0.986	0.170	0.668	0.693	0.988	0.110
1.732	1.472	0.223	0.638	0.448	1.0	0.500	0.772	0.25
2.0	1.70	0.135	0.446	0.406	1.155	0.428	0.653	0.236
3.0	2.54	0.0112	0.055	0.0896	1.732	0.250	0.333	0.125
4.0	3.39	0.0003	0.002	0.0068	2.31	0.158	0.177	0.059
5.0	4.23	0.00001	0.00003	0.0001	2.89	0.107	0.102	0.0295
6.0	5.08	...	...	...	3.46	0.077	0.063	0.016
7.0	5.94	...	...	...	4.04	0.058	0.041	0.01
8.0	6.78	...	...	...	4.63	0.045	0.0285	...

or line widths between slope extremes, and they are indicated here by Δ. They allow one to write readily the analytical expressions associated with these two line shape functions in terms of either of the two half-widths. This is done in Table 1, and since in magnetic resonance the peak-to-peak line width is more useful, all the expressions, except for the zero-derivative curve, are given in terms of Δ. Figure 1*a* indicates the peak-to-peak half-line widths Δ as well as the half-line width at half-maximum intensity δ for the two line shape functions.

If two curves have the same δ yet their peak-to-peak line widths do not coincide, obviously the reverse is true also. Figure 1*b* is a plot of the Gaussian and Lorentzian functions with the same peak-to-peak line width. Again the significant differences between the two line shape functions are made evident, especially the considerably slower rate at which the Lorentzian falls off in the wings.

In addition to the expressions for the absorption curves, Table 1 gives the expressions for the first, $F'(H - H_0)$, and second, $F''(H - H_0)$, derivatives as well as the maximum amplitude, $F(H - H_0)_{max}$, of the absorption, the amplitude of the first derivative at δ, and the amplitude of the maxima or inflection points for each curve. The values of $(H - H_0)$ at which each curve has maximum or minimum or inflection points are, of course, readily obtained by differentiating the original expression and then setting it equal to zero. Thus, both absorption curves have their maximum value at $(H - H_0)$ equal to zero, and they approach zero amplitude at large values of $(H - H_0)$, although the Gaussian drops toward zero much more rapidly than the Lorentzian (Fig. 1).

The first derivative curves (Fig. 2) again show their maximum (or minimum) value at $(H - H_0)$ equal to $\pm\Delta$, and fall off in the wings in a similar manner as the absorption curves. This difference in the rate of fall of slope in the wings of the two curves can be used to great advantage for a quick and unambiguous determination of the analytical expression a recorded curve fits. The second derivatives (Fig. 3) of both curves have solutions at the same points, namely at 0, $\pm\sqrt{3}\Delta$ and $\pm\infty$. However, the amplitude of these curves at the indicated points varies considerably. In fact the ratio of the amplitude at zero to the amplitude at $\pm\sqrt{3}\Delta$ is approximately 4 for a Lorentzian and approximately 2 for a Gaussian, and this fact can be used as a quick check as to the nature of an experimental curve.

So far a number of expressions have been discussed that allow one to ascertain quickly and unambiguously the type of curve that is obtained experimentally. This can be done regardless of whether one has the absorption curve itself or one of the derivatives.

One of the most useful pieces of information obtainable from the nmr or esr spectra is the number of unpaired spins that give rise to a particular signal.

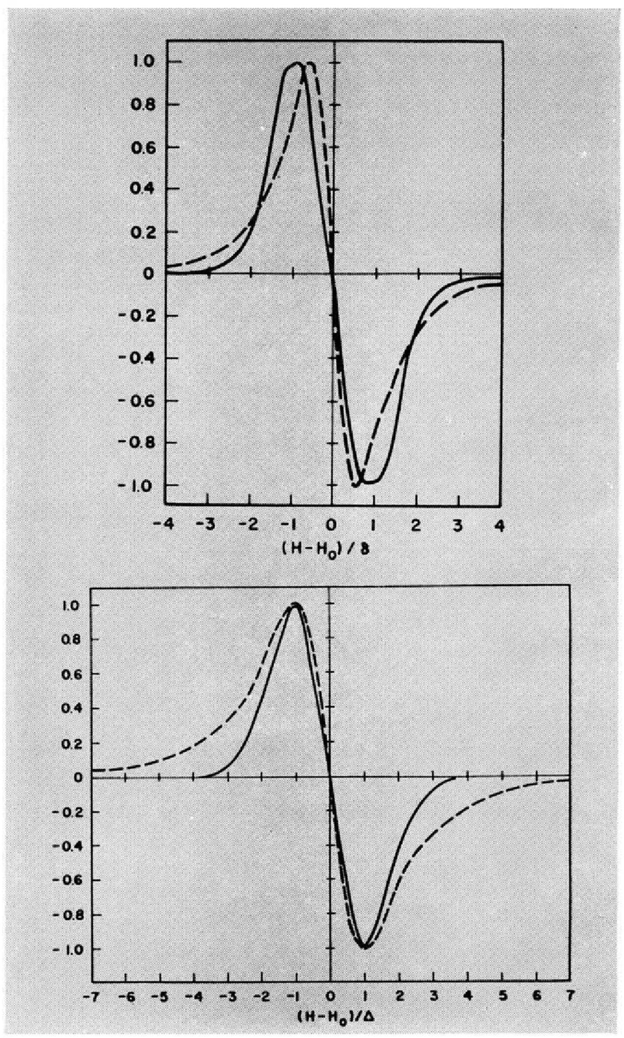

Figure 2. First derivative curves for Gaussian (solid line) and Lorentzian (broken line) shape function; (a) curves have the same half-line width at half-maximum intensity; (b) curves have the same peak-to-peak line width.

Unlike other spectroscopic techniques where one has to consider molar extinction coefficients, in magnetic resonance the area under a particular absorption is directly proportional to the number of species responsible for that signal. For example, in high resolution nmr if the ethyl group is present the ratio of the methyl area to that of the methylene is always 3/2. The integration, of course, of an absorption peak can be done electronically, graphically, etc., or alternately one can obtain the area directly from the line width and the amplitude. For the absorption curve the area is proportional to the product of the maximum amplitude and the line width at half maximum intensity,

$$K(2\delta)\mathrm{F}(H - H_0)_{max} \tag{6}$$

Substitution of the appropriate values for the normalized curves yields the area proportionality constants 1.062 for the Gaussian, and 1.572 for the Lorentzian, respectively.

If the first derivatives are available then the area is proportional to the product of maximum amplitude and the square of the line width between slope extrema,

$$K'2\mathrm{F}(H - H_0)_{max}(2\Delta)^2 \tag{7}$$

The respective proportionality constants are 0.517 for a Gaussian and 1.812 for a Lorentzian.

The foregoing indicate that caution must be exercised in comparing areas for different samples or even for a given sample recorded under different conditions or where a curve changes progressively from one function to another (*7*), and one must first ascertain the nature of the experimental curves obtained. Thus for two absorption curves, one Lorentzian and the other Gaussian, of the same line width at half-maximum intensity and of the same maximum intensity, the Lorentzian will have *1.48* as high a concentration as the Gaussian. If one compares the first derivative of two curves of the same peak-to-peak line width and of the same maximum amplitude, the Lorentzian will have *3.51* times as many spins as the corresponding Gaussian.

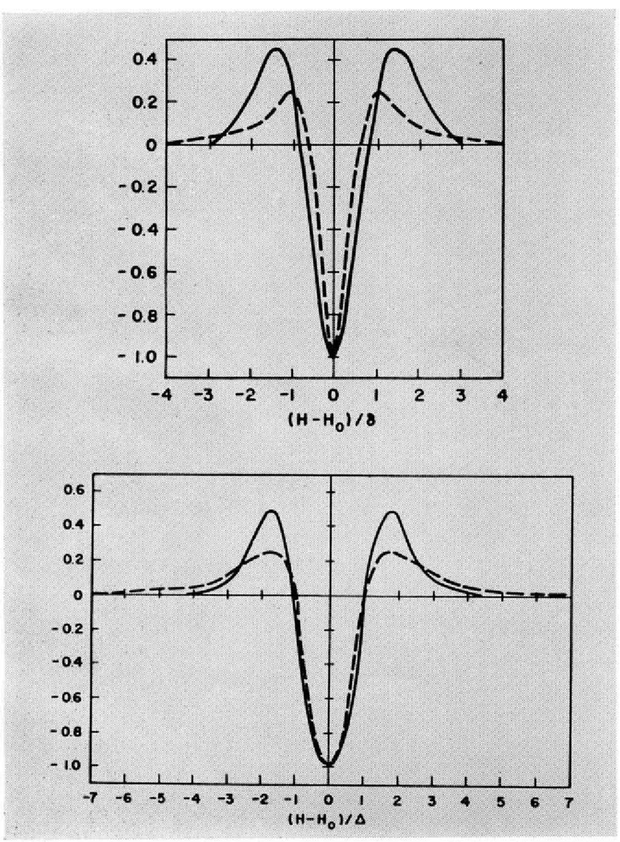

Figure 3. Second derivative curves for Gaussian (solid line) and Lorentzian shape function; (a) curves have the same half-line width at half-maximum intensity; (b) curves have the same peak-to-peak line width.

Spectral Moments

In addition to the spectral parameters discussed, i.e., the areas, line widths, and amplitudes, we will discuss briefly the moments of these functions also. Following Van Vleck's important paper (*8*) on the moments of nmr spectra, considerable attention has been given to the calculation of spectral moments especially in wide-line nmr. The nth moment of a curve is a very useful statistical parameter, and it is defined by

$$M_n = \int_0^\infty (H - H_0)^n \mathrm{F}(H - H_0)\mathrm{d}H \tag{8}$$

The nth moment of the corresponding first derivative curve is obtained by simple integration by parts of the above equation resulting in

$$M_n = -\frac{1}{n+1}\int_0^\infty (H - H_0)^{n+1}\mathrm{F}'(H - H_0)\mathrm{d}H \tag{9}$$

The moments of an experimental curve may be calculated readily by graphical or numerical methods. For the two line shapes here under consideration all odd moments vanish by symmetry. For the Lorentzian the second and higher even moments are infinite because for large $(H - H_0)$ the integrals behave as

$$\int_0^\infty (H - H_0)^{n-2}\,\mathrm{d}H \;\; (n = 2, 4, 6 \ldots)$$

On the other hand the even moments of the Gaussian are obtained readily from the expression

$$1{:}3{:}5 \ldots (n - 1)\Delta^n \qquad n = 2,4,6, \ldots \tag{10}$$

by virtue of the definite integral

$$\int x^n e^{-ax^2}\mathrm{d}x = \frac{1\cdot 3\cdot 5 \ldots (n - 1)}{2^{n/2+1}a^{n/2}}\sqrt{\pi/a}; \tag{11}$$

thus $M_2 = \Delta^2$, $M_4 = 3\Delta^4$, $M_6 = 5\Delta^6$.

The importance of the use of spectral moments in magnetic resonance spectroscopy has been demonstrated adequately. In high resolution nmr the moments have been used successfully in the analysis of complex spectra of multispin systems (*9*).

However, it is in wide-line nmr that spectral moments have found their greatest application. The exact analysis of wide-line nmr spectral patterns of more than three spin systems, either in single crystals or in polycrystalline materials, is prohibitive; but the calculation of the experimental spectral moments is very simple. Theoretical moments may be easily calculated for postulated arrangements of interacting spins from known or reasonable internuclear vectors, and angles that the internuclear vectors make with the magnetic field. Thus, by comparing the experimental and theoretical moments for the system under consideration one may be able to decide among alternate structures. Molecular and crystal structure parameters have been obtained in this manner as well as information about molecular motions such as diffusion and rotational transitions in the solid states (*2, 10*). The interested reader is referred to these indicated references, especially Abragam's treatment of the subject (*2*).

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